

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 4

6. QUANTUM MECHANICS II

• **Heisenberg uncertainty principle for the ground-state of infinite square-well potential**

The ground-state ($n = 1$) wave function of the infinite square-well potential of width L is

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}x\right).$$

Calculate $\langle x \rangle$, $\langle x^2 \rangle$, and the variance $(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$ for the position operator. Calculate the corresponding quantities for the momentum operator, i.e., $\langle p \rangle$, $\langle p^2 \rangle$, and the variance $(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2$. Verify that the Heisenberg uncertainty principle is fulfilled.

Hint: use the following definition for the momentum operator

$$p = \frac{\hbar}{i} \frac{d}{dx}.$$

General strategy

In general, the expectation value of an operator O on a quantum state $\psi(x)$ is given by

$$\langle O \rangle = \frac{\int_{-\infty}^{\infty} dx \psi^*(x) O \psi(x)}{\int_{-\infty}^{\infty} dx |\psi(x)|^2},$$

where the denominator takes care of the normalization.

Since the function $\psi_1(x)$ for the ground-state is properly normalized, calculating the expectation values for the position operator boil down to evaluating an integral of the following form

$$\langle x^n \rangle = \int_0^L dx \psi^*(x) x^n \psi(x).$$

The strategy for this integral is the following

- (i) use the trigonometric identity $\sin^2(\alpha) = \frac{1}{2} (1 - \cos(2\alpha))$, this way the first term of the integral is trivial
- (ii) perform the variable change $2\pi x/L = y$. to end up with an integral of the form

$$\left(\frac{L}{2\pi}\right)^{n+1} \int_0^{2\pi} dy y^n \cos(y)$$

- (iii) perform the integral by part using the following

$$\int u dv = uv - \int v du,$$

with $u = y^n$ and $dv = \cos(y)dy$.

- (iv) repeat step (iii) as many times as needed.

Position integrals

The first integral is,

$$\langle x \rangle = \frac{2}{L} \int_0^L dx x \sin^2\left(\frac{\pi}{L}x\right) = \frac{L}{2} - \frac{L^2}{4\pi} \left[y \sin(y) \right]_0^{2\pi} - \frac{L^2}{4\pi} \left[\int_0^{2\pi} dy \sin(y) \right] = \frac{L}{2},$$

where we note that the last two terms are both zero.

The second integral is

$$\langle x^2 \rangle = \frac{2}{L} \int_0^L dx x^2 \sin^2\left(\frac{\pi}{L}x\right) = L^2 \left(\frac{1}{3} - \frac{1}{2\pi^2} \right),$$

where one has to integrate by parts twice, and one of the terms will be zero.

Finally, the variance is given by

$$(\Delta x)^2 = L^2 \left(\frac{\pi^2 - 6}{12\pi^2} \right).$$

Momentum integrals

The first integral is,

$$\langle p \rangle = \frac{\hbar}{i} \frac{2}{L} \int_0^L dx \sin\left(\frac{\pi}{L}x\right) \frac{d}{dx} \sin\left(\frac{\pi}{L}x\right) = \frac{\hbar}{i} \frac{2}{\pi} \int_0^L dx \sin\left(\frac{\pi}{L}x\right) \cos\left(\frac{\pi}{L}x\right) = -\frac{\hbar}{i} \frac{L}{2\pi^2} \left[\cos(y) \right]_0^{2\pi} = 0,$$

which is zero by symmetry, but it can be verified explicitly by using the variable transformation and finding the primitive function.

The second integral is

$$\langle p^2 \rangle = -\hbar^2 \frac{2}{L} \int_0^L dx \sin\left(\frac{\pi}{L}x\right) \frac{d^2}{dx^2} \sin\left(\frac{\pi}{L}x\right) = \hbar^2 \frac{\pi^2}{L^2} \frac{2}{L} \int_0^L dx \sin^2\left(\frac{\pi}{L}x\right) = \frac{\hbar^2 \pi^2}{L^2},$$

where, after performing the differentiation, one could realize it is the same integral that gives the normalization condition of the wave function, aside from a constant prefactor.

Finally, the variance is given by

$$\Delta p = \frac{\hbar \pi}{L}$$

Heisenberg uncertainty principle

For the ground state $\psi_1(x)$, the Heisenberg uncertainty principle reads

$$\Delta x \Delta p = \sqrt{\frac{\pi^2 - 6}{12}} \hbar \approx 0.568 \hbar > \frac{\hbar}{2}.$$

It can be verified that this condition is verified for any wavefunction $\psi_n(x)$ with $n \geq 1$, for which one gets a position uncertainty

$$\Delta x = \frac{L}{2\pi n} \sqrt{\frac{\pi^2}{3} n^2 - 2}$$

and a momentum uncertainty

$$\Delta p = \frac{\pi \hbar}{L} n$$

thus yielding

$$\Delta x \Delta p = \frac{\hbar}{2} \sqrt{\frac{\pi^2}{3} n^2 - 2} > \frac{\hbar}{2} \quad \forall n \in \mathbb{N}/\{0\}$$

• **Heisenberg uncertainty principle in the nucleus / Ex. 5.37**

A neutron is confined in a deuterium nucleus (deuteron) of diameter $d = 3.1 \times 10^{-15}$ m. Use the energy-level calculation of a one-dimensional box to calculate the neutron's minimum kinetic energy. What is the neutron's minimum kinetic energy according to the uncertainty principle?

Solution

For a box of the size of the deuteron, the energy levels are given by

$$E_n = \frac{h^2}{8m_n d^2} n^2,$$

so the lowest level ($n = 1$) corresponds to

$$E_1 = \frac{h^2}{8m_n d^2} = \frac{(hc)^2}{8(m_n c^2) d^2} = 21.3 \text{ MeV},$$

where $m_n c^2 = 939.57 \text{ MeV}$ and $hc = 1240 \text{ eV nm}$.

By using the uncertainty principle, the minimum of the kinetic energy corresponds to the minimum of the momentum, so

$$\Delta p \Delta x = \frac{\hbar}{2},$$

and for $\Delta x = d$, i.e., the neutron is confined in the nucleus

$$E_{\min} = \frac{(\Delta p)^2}{2m} = \left(\frac{\hbar}{2d} \right)^2 \frac{1}{2m} = \frac{\hbar^2}{8md^2},$$

which differs from the estimate obtained for a particle in a box by the 2π factor(s) coming from $\hbar$ vs h .

• **Heisenberg uncertainty principle for the quantum harmonic oscillator / Examples 6.12 & 6.13**

Normalize the ground-state wave function $\psi_0(x)$ for the harmonic oscillator and verify that the Heisenberg uncertainty principle $\Delta x \Delta p \geq \hbar/2$ is verified.

Solution

For the ground state, the wave function of the quantum harmonic oscillator has the Gaussian form

$$\psi_0(x) = A e^{-\alpha x^2/2},$$

where the normalization constant A can be found as

$$1 = |A|^2 \int_{-\infty}^{+\infty} dx \psi^*(x) \psi(x) = |A|^2 \int_{-\infty}^{+\infty} dx e^{-\alpha x^2} = 2|A|^2 \int_0^{+\infty} dx e^{-\alpha x^2} = 2|A|^2 \left(\frac{1}{2} \sqrt{\frac{\pi}{\alpha}} \right)$$

thus yielding the following normalized ground-state wave function

$$\psi_0(x) = \left(\frac{\alpha}{\pi} \right)^{1/4} e^{-\alpha x^2/2}.$$

In order to verify the Heisenberg uncertainty principle, one needs to evaluate a series of integrals, corresponding to the expectation values $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, and $\langle p^2 \rangle$.

Therefore, for the position

$$\langle x \rangle = \int_{-\infty}^{+\infty} dx \psi^*(x) x \psi(x) = \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} dx x e^{-\alpha x^2} = 0$$

which is zero by symmetry (integral of an odd function on a symmetric domain) whereas

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} dx \psi^*(x) x^2 \psi(x) = \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} dx x^2 e^{-\alpha x^2} = 2\sqrt{\frac{\alpha}{\pi}} \int_0^{+\infty} dx x^2 e^{-\alpha x^2} = 2\sqrt{\frac{\alpha}{\pi}} \left(\frac{\sqrt{\pi}}{4\alpha^{3/2}} \right) = \frac{1}{2\alpha} = \frac{\hbar}{2m\omega}$$

where $\omega = \sqrt{\kappa/m}$ is the characteristic frequency of the oscillator, analogously to the classical case. Hence, the uncertainty for the position is given by

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{2m\omega}}$$

For the momentum, recalling that $p = -i\hbar \frac{d}{dx}$

$$\langle p \rangle = -i\hbar \int_{-\infty}^{+\infty} dx \psi^*(x) \frac{d}{dx} \psi(x) = i\alpha\hbar \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} dx x e^{-\alpha x^2} = 0$$

by the same argument as above, whereas

$$\langle p^2 \rangle = -\hbar^2 \int_{-\infty}^{+\infty} dx \psi^*(x) \frac{d^2}{dx^2} \psi(x) = -\alpha\hbar^2 \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} dx (\alpha x^2 - 1) e^{-\alpha x^2} = \frac{\alpha\hbar^2}{2} = \frac{\hbar m\omega}{2}.$$

Hence, the uncertainty for the momentum is given by

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{\hbar m\omega}{2}}$$

Finally, the Heisenberg uncertainty principle reads

$$\Delta x \Delta p = \frac{\hbar}{2} \geq \frac{\hbar}{2}$$

and corresponds to the lowest possible uncertainty.

Observation: This is related to the property of the Gaussian function to remain Gaussian under Fourier transform. Recall that Fourier transform exchanges $-i\frac{\partial}{\partial x}$ (which is proportional to the momentum operator) with k (the new wavenumber variable). Hence Fourier transforming the wave function it is possible to compute the momentum variance using the same formulas as the position variance. The variance of a Gaussian function in the Fourier domain is proportional to the inverse of the variance in the direct domain. Therefore, if the function spreads in one of the domains, it shrinks in the other, thus every Gaussian regardless of its width minimizes the product of the variances. The reason this solution appears for the harmonic oscillator is also interesting, the Harmonic oscillator Hamiltonian retains its form under Fourier transformation, just the role of the kinetic and potential terms is switched.

• **Gaussian integrals and useful relations**

Basic integrals. Consider the integral of the form

$$I_n(\alpha) = \int_0^\infty dx x^n e^{-\alpha x^2}.$$

For $n = 0$ one obtains the usual Gaussian integral

$$I_0(\alpha) = \int_0^\infty dx e^{-\alpha x^2} = \frac{\sqrt{\pi}}{2} \alpha^{-1/2},$$

which can be solved by squaring it and then solving it in polar coordinates. Instead, for $n = 1$

$$I_1(\alpha) = \int_0^\infty dx x e^{-\alpha x^2} = \frac{1}{2} \alpha^{-1},$$

which can be solved with the substitution $u = \alpha x^2$ so that $du = 2\alpha x dx$, leaving a simple integral.

Recursive formula. Note that taking the derivative with respect to α on both sides

$$I_2(\alpha) = -\frac{\partial}{\partial \alpha} I_0(\alpha) = \int_0^\infty dx x^2 e^{-\alpha x^2} = \frac{\sqrt{\pi}}{4} \alpha^{-3/2}.$$

This can be generalized to $n > 1$, obtaining the recursive formula

$$I_n(\alpha) = -\frac{\partial}{\partial \alpha} I_{n-2}(\alpha),$$

which for $n = 2m$ (even) yields

$$I_n(\alpha) = -\frac{\partial}{\partial \alpha} I_{n-2}(\alpha) = \dots = \frac{\partial^{n/2}}{\partial \alpha^{n/2}} I_0(\alpha) = \frac{\sqrt{\pi}}{2} \frac{\partial^{n/2}}{\partial \alpha^{n/2}} \alpha^{-1/2}.$$

and for $n = 2m + 1$ (odd) yields

$$I_n(\alpha) = -\frac{\partial}{\partial \alpha} I_{n-2}(\alpha) = \dots = \frac{\partial^{(n-1)/2}}{\partial \alpha^{(n-1)/2}} I_1(\alpha) = \frac{1}{2} \frac{\partial^{(n-1)/2}}{\partial \alpha^{(n-1)/2}} \alpha^{-1}.$$

Other integrals. The related family of integrals

$$H_n(\alpha) = \int_{-\infty}^\infty dx x^n e^{-\alpha x^2}$$

have solutions

$$H_n(\alpha) = \begin{cases} 2I_n(\alpha), & \text{if } n = 2m \\ 0, & \text{if } n = 2m + 1 \end{cases}$$

• Ehrenfest theorem

An important property of quantum mechanics is that it gives back the familiar results of classical mechanics in the limit of macroscopic objects. We already saw for the Heisenberg uncertainty relations, that the typical uncertainties of position and momentum for macroscopic objects are so small, that they are practically negligible. Now we show that the *expectation values* of the observables obey classical mechanics.

The Ehrenfest theorem is a general statement, here we look at the specific case relating position and momentum:

$$m \frac{d}{dt} \langle \hat{x} \rangle = \langle \hat{p} \rangle.$$

This relation is true for expectation values taken in *any* quantum state (as long as the position and momentum expectation values are well-defined). Identifying the expectation values with classical variables, this is identical to the classical relation of velocity and momentum:

$$m \frac{dx}{dt} = p.$$

Prove the Ehrenfest theorem by expanding the left hand side and using the time-dependent Schrödinger equation.

Proof

To prove this relation we expand the derivative of the expectation value using the linearity of integrals and the chain rule:

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{d}{dt} \int_{-\infty}^{\infty} dx \psi^*(x, t) x \psi(x, t) = \int_{-\infty}^{\infty} dx \left(\frac{\partial \psi^*}{\partial t}(x, t) x \psi(x, t) + \psi^*(x, t) x \frac{\partial \psi}{\partial t}(x, t) \right).$$

Here, we only consider one-dimensional systems, but the derivation works the same way in higher dimensions, as terms containing the other coordinate variables all cancel in the following steps. In the following, we drop the explicit dependence on x and t for brevity.

In the next step, we use the Schrödinger equation to express the time derivative:

$$\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar} \hat{H} \psi = \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi.$$

Taking the complex conjugate of the equation, we also get

$$\frac{\partial \psi^*}{\partial t} = -\frac{1}{i\hbar} \hat{H}^* \psi^* = -\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi^*.$$

Substituting these back into the formula for $\frac{d}{dt} \langle \hat{x} \rangle$, we find that the terms containing $V(x)$ cancel, and we are left with

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{i\hbar}{2m} \int_{-\infty}^{\infty} dx \left(\psi^* x \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} x \psi \right).$$

Next we integrate by parts in both terms:

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{i\hbar}{2m} \left[\psi^* x \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} x \psi \right]_{-\infty}^{\infty} - \frac{i\hbar}{2m} \int_{-\infty}^{\infty} dx \left\{ \frac{\partial}{\partial x} (\psi^* x) \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \frac{\partial}{\partial x} (x \psi) \right\}.$$

The first term vanishes because if ψ is a normalizable wavefunction with well-defined position and momentum expectation values, it has to decay quickly in infinity. In the second term, we expand the derivatives, and after cancellations, we find

$$\frac{d}{dt} \langle \hat{x} \rangle = -\frac{i\hbar}{2m} \int_{-\infty}^{\infty} dx \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right).$$

We integrate the second term by parts again and use the vanishing of the wavefunction in infinity:

$$\frac{d}{dt} \langle \hat{x} \rangle = -\frac{i\hbar}{2m} \left\{ \int_{-\infty}^{\infty} dx \left(\psi^* \frac{\partial \psi}{\partial x} + \psi^* \frac{\partial \psi}{\partial x} \right) - [\psi^* \psi]_{-\infty}^{\infty} \right\} = -\frac{i\hbar}{m} \int_{-\infty}^{\infty} dx \psi^* \frac{\partial \psi}{\partial x} = \frac{1}{m} \langle \hat{p} \rangle,$$

where we used that the momentum operator is

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}.$$

□

Remark: It is also possible to reproduce Hamilton's other equation for the expectation values using a similar calculation:

$$\frac{d}{dt} \langle \hat{p} \rangle = - \left\langle \frac{\partial V}{\partial x}(\hat{x}) \right\rangle,$$

where the left hand side is the rate of change of momentum, and the right hand side is the force.

• **Harmonic oscillator approximation of the H₂ molecule / Ex. 6.39**

An H₂ molecule can be approximated by a simple harmonic oscillator with a force constant $\kappa = 1.1 \times 10^3$ N/m. Find (a) the energy levels E_n and (b) the possible wavelengths λ of photons emitted when the H₂ molecule decays from the third excited state eventually to the ground state.

Solution

(a) Each H atom is composed of a single proton with mass m_p . One can reduce the two-body problem to the problem of a single body oscillating against a fixed point. As both protons move in such a way to satisfy the conservation of momentum (i.e., keeping the center of mass stationary), the system behaves as if its mass were only half of the mass of a single proton. Hence, the corresponding harmonic frequency is given by

$$\omega = \sqrt{\frac{\kappa}{\mu}} = \sqrt{\kappa \left(\frac{m_1 + m_2}{m_1 m_2} \right)} = \sqrt{\kappa \frac{2}{m_p}}$$

and the energies of the harmonic oscillators are given by

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right).$$

(b) There are a series of possible transitions from the $n = 3$ state to the ground state $n = 0$, and the energy of the emitted photon is given by

$$E_\gamma = \Delta n \hbar\omega$$

where $\Delta n = n_i - n_f$ is the difference between the integer labeling the initial and final states.

Using $\hbar = 6.582 \times 10^{-16}$ eV s and $m_p = 1.673 \times 10^{-27}$ kg one finds

- $\hbar\omega = 0.755$ eV
- $2\hbar\omega = 1.51$ eV
- $3\hbar\omega = 2.26$ eV

corresponding to a photon wavelength are evaluated as $\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{E}$, yielding

- $\lambda = 1640$ nm
- $\lambda = 821$ nm
- $\lambda = 549$ nm

for the corresponding transition energies.

• **Classical vs. quantum harmonic oscillators / Ex. 6.59**

In the lab you make a simple harmonic oscillator with a mass $m = 0.15 \text{ kg}$ attached to a spring with elastic constant $\kappa = 12 \text{ N m}^{-1}$. (a) If the oscillation amplitude is $A = 0.10 \text{ m}$, what is the corresponding quantum number n for the quantum harmonic oscillator? (b) What would be the amplitude of the quantum ground state for this oscillator? (c) What is the energy of a photon emitted when this oscillator makes a transition between adjacent energy levels? Comment on each of your results.

Solution

(a) The energy of the classical harmonic oscillator is given by

$$E = \frac{1}{2} \kappa A^2$$

in the configuration of *maximum elongation*, in which the mass velocity (and hence its kinetic energy) is zero. Comparing the classical and quantum energy

$$\hbar \omega \left(n + \frac{1}{2} \right) = \hbar \sqrt{\frac{\kappa}{m}} \left(n + \frac{1}{2} \right) = \frac{1}{2} \kappa A^2$$

where the classical frequency is $\omega = \sqrt{\frac{\kappa}{m}}$. The corresponding value of n is given by

$$n + \frac{1}{2} = \frac{A^2 \sqrt{\kappa m}}{2\hbar} = 6.4 \times 10^{31}$$

because the classical system is exceedingly large as compared to a quantum oscillator.

(b) for the ground state $n = 0$, the corresponding amplitude is given by

$$A = \left(\frac{\hbar^2}{\kappa m} \right)^{1/4} = 8.9 \times 10^{-18} \text{ m},$$

which is *eight* orders of magnitude smaller than Bohr's radius.

(c) Finally, the photon energy emitted for transitions with $\Delta n = 1$ is given by

$$E_\gamma = \hbar \omega = \hbar \sqrt{\frac{\kappa}{m}} = 9.4 \times 10^{-34} \text{ J},$$

which is too small to be observed.

• Attractive Dirac delta potential

Find the bound state ($E < 0$) energy and wave function of the 1D Schrödinger equation with potential

$$V(x) = -\alpha \delta(x),$$

where $\alpha > 0$ is a constant with eV m units (why?). Use the identity that the derivative of the sign function is a Dirac delta $\frac{d}{dx} \text{sign}(x) = 2\delta(x)$. This is used to model a strong and localized attractive potential.

Solution

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - \alpha \delta(x) \psi(x) = E \psi(x).$$

We need to find a wavefunction ψ that solves this differential equation for some $E < 0$.

Let us first find the solutions away from $x = 0$ where the potential term vanishes. Here, the equation reduces to

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E \psi(x),$$

and we can look for solutions in the form

$$\psi(x) = e^{\lambda x}.$$

Note that we didn't include a prefactor, as for the Schrödinger equation it appears on both sides and can be canceled. We can take care of properly normalizing the wavefunction at the very end if necessary. Substituting back, we find

$$-\frac{\hbar^2}{2m} \lambda^2 = E \Rightarrow \lambda_{\pm} = \pm \frac{\sqrt{-2mE}}{\hbar} = \pm \lambda.$$

Here we use the fact that $E < 0$, so the solutions $\lambda_{\pm}$ are real, describing exponentially increasing/decreasing wavefunctions. Neither of these solutions is normalizable by itself. However, so far we have only considered what happens away from $x = 0$. At $x = 0$, the potential is nonzero, and we can stitch the solutions together to get a normalizable wavefunction:

$$\psi(x) = e^{-\lambda|x|} = \begin{cases} e^{\lambda-x}, & \text{if } x > 0, \\ e^{\lambda+x}, & \text{if } x < 0. \end{cases}$$

This wavefunction is continuous, decays exponentially for $x \rightarrow \pm\infty$, and its square is integrable over the real line.

The final step is to find the value of λ (hence E) which solves the Schrödinger equation at $x = 0$. For this, we evaluate the kinetic term on the trial wavefunction above:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} e^{-\lambda|x|} = \frac{\hbar^2}{2m} \frac{\partial}{\partial x} \left(\lambda \text{sign}(x) e^{-\lambda|x|} \right) = -\frac{\hbar^2}{2m} (-2\lambda\delta(x) + \lambda^2) e^{-\lambda|x|},$$

where we used that $\frac{d}{dx}|x| = \text{sign}(x)$ and $\text{sign}(x)^2 = 1$, together with the identity $\frac{d}{dx} \text{sign}(x) = 2\delta(x)$.

We substitute this expression in the time-independent Schrödinger equation and get

$$-\frac{\hbar^2}{2m} (-2\lambda\delta(x) + \lambda^2) e^{-\lambda|x|} - \alpha \delta(x) e^{-\lambda|x|} = E e^{-\lambda|x|}.$$

We order the equation to zero, and group the terms depending on whether they include or not a Dirac delta. For the terms without a Dirac delta we find the relation

$$-\frac{\hbar^2}{2m} \lambda^2 = E,$$

which is the condition we already obtained away from $x = 0$. For the terms with a Dirac delta, we find

$$\lambda \frac{\hbar^2}{m} - \alpha = 0 \Rightarrow \lambda = \frac{\alpha m}{\hbar^2}$$

By combining them, we obtain an expression for the energy of the *bound state*

$$E = -\frac{m\alpha^2}{2\hbar^2}.$$

Finally, we can verify by dimensional analysis that α has dimension [eV m].